Least singular values of independent random tensor columns

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Abstract

Let A have $m = \lfloor (1-\varepsilon)n^d \rfloor$ independent columns, each the tensor product of d random vectors in $\mathbb{R}^n$. All scalar coordinates are independent, centered, of variance one, and have subgaussian norm at most K. We prove that, for every fixed $\varepsilon \in (0, 1/2]$ and every integer sequence $d = o(\sqrt{n/\log n})$, the least singular value is at least $c_K \varepsilon n^{d/2}$ with probability tending to one. The constant depends only on K; neither identical distributions nor symmetry of the coordinates is required. The proof conditions on all but the last tensor factor, estimates quadratic Gram moments of projected column blocks, and lifts the block estimate by averaging over subsets of columns.

1 Introduction

Independent rank-one tensor features arise when coefficients are recovered from a linear combination of simple tensors. If these features are the columns of a matrix A, a lower bound for its least singular value controls the amplification of an error in the observed linear combination. Tensor columns are isotropic under the assumptions below, but their scalar entries are dependent. The tensor order may also grow with the factor dimension.

For a real random variable ξ, use the convention

$$\|\xi\|_{\psi_2} = \inf\{s > 0 : \mathbb{E}\exp(\xi^2/s^2) \le 2\}.$$

Let $x_{i\ell} \in \mathbb{R}^n$, $1 \le i \le m$, $1 \le \ell \le d$. Assume that all coordinates of all these vectors are mutually independent, with mean zero, variance one, and subgaussian norm at most K. Define

$$X_i = x_{i1} \otimes \cdots \otimes x_{id}, \qquad A = [X_1, \dots, X_m] \in \mathbb{R}^{N \times m}, \qquad N = n^d.$$

The coordinates can have different laws, and the laws can depend on n. For a matrix with $m \ge 1$ columns, write

$$s_{\min}(A) = \inf\{\|Av\| : v \in \mathbb{R}^m,\ \|v\| = 1\}.$$

Vershynin's tensor concentration theorem has constants independent of the order and yields quantitative linear independence [1, Theorem 1.4 and Corollary 6.2]. In the regime $C_K d^2 \log(n)/n \le \varepsilon \le 1/2$, that corollary gives $s_{\min}(A) \ge \sqrt{\varepsilon}/2$ with probability at least $1 - 2\exp(-c_K \varepsilon n/d)$, when $m \le (1-\varepsilon)n^d$. Section 1.5 of the same paper asks for an improvement to the ambient dimension scale. The following theorem supplies that scale with a constant independent of the tensor order.

Theorem 1. *For every $K > 0$ there is $c_K > 0$ with the following property. Fix any $\varepsilon \in (0, 1/2]$, and let $d = d(n) \geq 1$ be an integer sequence such that*

$$\frac{d^2 \log n}{n} \longrightarrow 0.$$

Put $N = n^d$ and $m = \lfloor (1-\varepsilon)N \rfloor$. For every sequence of coordinate laws satisfying the independence, centering, variance and subgaussian assumptions above,

$$\mathbb{P}\{s_{\min}(A) \geq c_K \varepsilon \sqrt{N}\} \longrightarrow 1 \qquad (n \longrightarrow \infty).$$

The constant c_K is independent of n, d, ε and the coordinate laws. Probability convergence is asserted for each fixed ε.

This is the full target of Problem 562 in the AIM collection [5], at the revision specified in that reference.

An estimate of the distance of each column from the span of the others does not by itself give this scale: selecting a largest coordinate of a unit coefficient vector can lose a factor $\sqrt{m}$. We instead project a small block of k columns away from all remaining columns. Its Gram matrix is well conditioned at the projection-rank scale. Once this holds with high probability for each fixed block, a deterministic subset-averaging argument gives the conclusion for all coefficient vectors without a union bound over all blocks.

To estimate a projected block, we condition on its first $d-1$ factors. The remaining Gram fluctuation is a quadratic polynomial in independent scalars. A scalar convex comparison reduces its off-diagonal part to a Gaussian quadratic polynomial with constants depending only on K. We then prove a Gaussian Gram moment bound using Hermite creation and annihilation operators. This use of low-degree operator norms follows the method of Bandeira, Kunisky, Nizić-Nikolac, Pesenti and Wang [3, Section 2]; the coefficient identities and estimates needed here are proved in Section 2. Tensor concentration [1] and matrix Chernoff [2] control the conditioned coefficients. These are the two concentration estimates imported into the proof.

All vector norms are Euclidean and all matrix norms are induced operator norms. The Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\mathrm{HS}}$, and Tr is the unnormalized trace. For a square matrix H, $\mathrm{off}(H) = H - \mathrm{diag}(H)$. All logarithms are natural.

2 Quadratic Gram moments

2.1 Scalar comparison

We first record an elementary scalar form of convex domination. For broader vector comparison results, see van Handel [4].

Lemma 2. *If $\mathbb{E}\xi = 0$ and $\|\xi\|_{\psi_2} \leq K$, and g is standard normal, then*

$$\mathbb{E}f(\xi) \leq \mathbb{E}f(16Kg)$$

for every finite convex function $f : \mathbb{R} \to \mathbb{R}$ of polynomial growth.

Proof. Put $Y = \xi/(2K)$. Then $\mathbb{E}e^{Y^2} \leq 2$, so $\mathbb{P}\{|Y| > u\} \leq 2e^{-u^2}$. The inequality $Y^2 \leq e^{Y^2} - 1$ gives $\mathbb{E}Y^2 \leq 1$, and hence $\mathbb{E}|Y| \leq 1$. For $0 \leq z \leq 2$,

$$\mathbb{E}(Y - z)_+ \leq 1 < \frac{8}{\sqrt{2\pi}} - 2 \leq \mathbb{E}(8g - z)_+.$$

For $z \geq 2$, integration of the upper tail gives

$$\mathbb{E}(Y - z)_+ \leq 2 \int_z^\infty e^{-u^2}\, du \leq \frac{e^{-z^2}}{z} \leq e^{-z^2}.$$

On the event $g \geq (z+1)/8$, the positive part $(8g - z)_+$ is at least one. Integrating the Gaussian density over the interval $[(z+1)/8, (z+9)/8]$ therefore gives

$$\mathbb{E}(8g - z)_+ \geq \mathbb{P}\{g \geq (z+1)/8\} \geq \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(z+9)^2}{128}\right) \geq e^{-z^2}.$$

The last inequality holds because $z^2 - (z+9)^2/128$ is increasing for $z \geq 2$, and its value at 2 is greater than $3 > \log\sqrt{2\pi}$. The same bounds hold for $-Y$. Since Y is centered, $\mathbb{E}(Y-z)_+ = -z + \mathbb{E}(-Y+z)_+$; this proves the comparison of positive parts for negative z as well.

A finite convex piecewise affine function is an affine function plus a finite nonnegative linear combination of functions $(x - z)_+$. The comparison just proved and centering give its expectation inequality. For a general finite convex f, choose one supporting line at each rational point and include a fixed supporting line in every finite maximum. These maxima increase to f: supporting slopes are locally bounded by secant slopes, so rational points approaching any point recover its value. Subtracting the fixed line leaves a nonnegative increasing sequence. Monotone convergence proves the expectation comparison for Y and $8g$. Applying this to $f(2K\,\cdot)$ proves the claim. The polynomial-growth assumption ensures that all expectations are finite. □

Lemma 3. *Let $\xi_1, \ldots, \xi_s$ be independent centered scalars with $\|\xi_\alpha\|_{\psi_2} \leq K$. Let $H_{\alpha\lambda}$ be real symmetric matrices of the same size, with $H_{\alpha\lambda} = H_{\lambda\alpha}$ and $H_{\alpha\alpha} = 0$. For independent standard normals g_α, and any $q \geq 1$,*

$$\left(\mathbb{E}\left\|\sum_{\alpha,\lambda} H_{\alpha\lambda}\xi_\alpha\xi_\lambda\right\|^q\right)^{1/q} \leq 2(16K)^2 \left(\mathbb{E}\left\|\sum_{\alpha,\lambda} H_{\alpha\lambda}g_\alpha g_\lambda\right\|^q\right)^{1/q}.$$

Proof. Write $Q(x) = \sum_{\alpha,\lambda} H_{\alpha\lambda}x_\alpha x_\lambda$. Include each scalar index in a random subset S independently with probability $1/2$, and let

$$Q_S(x) = \sum_{\alpha\in S,\ \lambda\notin S} H_{\alpha\lambda}x_\alpha x_\lambda.$$

Then $Q(x) = 4\mathbb{E}_S Q_S(x)$, so convexity of the norm power yields

$$\|Q(\xi)\|_{L^q} \leq 4\big(\mathbb{E}_S\mathbb{E}\|Q_S(\xi)\|^q\big)^{1/q}.$$

For fixed S, the expression Q_S is affine in each separate coordinate. Conditional on the others, its norm to the power q is a convex function of that coordinate. Successive application of Lemma 2, followed by degree-two homogeneity, gives

$$\|Q_S(\xi)\|_{L^q} \leq (16K)^2\|Q_S(g)\|_{L^q}.$$

If g^S changes the signs of the coordinates indexed by S, symmetry of the coefficient array implies $Q_S(g) = (Q(g) - Q(g^S))/4$. Since g^S has the same law as g, the L^q triangle inequality gives $\|Q_S(g)\|_{L^q} \leq \|Q(g)\|_{L^q}/2$. Combining the bounds proves the lemma. □

2.2 Gaussian column matrices

The next estimate keeps both covariance scales separate. This is what makes a small projected block useful.

Lemma 4. *Let $U_i : \mathbb{R}^n \to \mathbb{R}^M$, $1 \le i \le k$, be deterministic real matrices, and let g_i be independent standard Gaussian vectors in $\mathbb{R}^n$. Put*

$$W = [U_1 g_1, \dots, U_k g_k], \quad D = \mathbb{E}W^T W = \operatorname{diag}(\|U_i\|_{\mathrm{HS}}^2), \quad L = \mathbb{E}WW^T = \sum_i U_i U_i^T,$$

and write $a_0 = \|D\|$, $b_0 = \|L\|$, and $t = \max_i \|U_i\|^2$. For every integer $p \ge 1$, set

$$\mathcal{M}_p = b_0 + 2(2p+1)t + 2\sqrt{(a_0 + (2p+1)t)(b_0 + (2p+1)t)}.$$

Then

$$\left(\mathbb{E}\|\operatorname{off}(W^T W)\|^{2p}\right)^{1/(2p)} \le 2k^{1/(2p)}\mathcal{M}_p. \tag{1}$$

Proof. Denote the columns of U_i by u_{ia}, index scalar coefficients by $\alpha = (i,a)$, and set $C_\alpha = u_{ia}e_i^T \in \mathbb{R}^{M\times k}$. Thus $W = \sum_\alpha g_\alpha C_\alpha$. We need the two self-adjoint block matrices

$$R_{\alpha\lambda} = C_\lambda^T C_\alpha, \qquad S_{\alpha\lambda} = C_\lambda C_\alpha^T. \tag{2}$$

They satisfy

$$\|R\| \le t, \qquad \|S\| \le t. \tag{3}$$

Indeed, for an input with coordinates $f_{(h,b);i}$, the output of R is

$$(Rf)_{(i,a);h} = \sum_b \langle u_{hb}, u_{ia}\rangle f_{(h,b);i}.$$

For each ordered pair (i,h), this is the map $U_i^T U_h$ from the input block labelled (h,i) to the output block labelled (i,h). Both block families are disjoint, and each map has norm at most t. Summing squared norms proves the first bound. For S, coefficients with distinct column indices vanish. For each i, the block matrix $[u_{ib}u_{ia}^T]_{a,b}$ is

$$(I_n \otimes U_i)F(I_n \otimes U_i^T),$$

where F interchanges the two $\mathbb{R}^n$ factors. This gives the second bound.

We use the Gaussian L^2 inner product on polynomials. The Hermite polynomials defined by $H_0 = 1$, $H_1(x) = x$, and $H_{\ell+1} = xH_\ell - \ell H_{\ell-1}$ satisfy $H_\ell' = \ell H_{\ell-1}$. For the standard Gaussian density ϕ, $\phi^{(\ell)} = (-1)^\ell \phi H_\ell$. Integration by parts ℓ times shows that $\mathbb{E}H_\ell(g)H_j(g) = \mathbb{E}H_j^{(\ell)}(g)$. This is zero if $\ell > j$, and equals $\ell!$ if $\ell = j$; the other unequal case follows by symmetry. The boundary terms vanish because they are polynomials times ϕ. The normalized polynomials $h_\ell = H_\ell/\sqrt{\ell!}$ therefore form an orthonormal basis of the polynomial space. Their recurrence is

$$xh_\ell(x) = \sqrt{\ell+1}\, h_{\ell+1}(x) + \sqrt{\ell}\, h_{\ell-1}(x).$$

For $\eta \in \mathbb{N}_0^{kn}$, the products $h_\eta = \prod_\alpha h_{\eta_\alpha}$ form the corresponding multivariate orthonormal basis. Define creation and annihilation on this basis by

$$c_\alpha h_\eta = \sqrt{\eta_\alpha + 1}\, h_{\eta + e_\alpha}, \qquad c_\alpha^* h_\eta = \sqrt{\eta_\alpha}\, h_{\eta - e_\alpha}.$$

A term with $\eta_\alpha = 0$ in the second formula is zero. Multiplication by g_α is $c_\alpha + c_\alpha^*$, and

$$c_\lambda^* c_\alpha - c_\alpha c_\lambda^* = \mathbf{1}_{\alpha=\lambda} I.$$

Let $\mathcal{P}_{\le q}$ be the polynomials of total degree at most q, and let $\Lambda f = (c_\alpha^* f)_\alpha$. On $\mathcal{P}_{\le q}$, also for vector-valued polynomials,

$$\|\Lambda\| \le \sqrt{q}, \tag{4}$$

since $\sum_\alpha c_\alpha c_\alpha^* h_\eta = |\eta| h_\eta$.

Define $\mathcal{U} = \sum_\alpha C_\alpha c_\alpha$ and $\mathcal{V} = \sum_\alpha C_\alpha c_\alpha^*$. These map k-component polynomials to M-component polynomials, and multiplication by W is $\mathcal{U} + \mathcal{V}$. The commutator and (2) give the identities

$$\mathcal{U}^*\mathcal{U} = D + \Lambda^* R \Lambda, \qquad \mathcal{V}\mathcal{V}^* = L + \Lambda^* S \Lambda. \tag{5}$$

Consequently, on degree at most q,

$$\|\mathcal{U}\| \le \sqrt{a_0 + qt}, \qquad \|\mathcal{V}^*\| \le \sqrt{b_0 + qt}.$$

On degree at most $q \ge 1$, their degree-lowering adjoints satisfy

$$\|\mathcal{U}^*\| \le \sqrt{a_0 + (q-1)t}, \qquad \|\mathcal{V}\| \le \sqrt{b_0 + (q-1)t}.$$

For these last bounds, restrict the raising maps to degree at most $q-1$ and take adjoints; homogeneous degrees are orthogonal. Both lowering maps vanish on constants. All norms here are restrictions between finite-dimensional polynomial spaces. No boundedness of these operators on the full Gaussian L^2 space is asserted.

Set $T = W^T W - D$, viewed also as a multiplication operator. By (5),

$$T = \Lambda^* R \Lambda + \mathcal{U}^*\mathcal{V} + \mathcal{V}^*\mathcal{U} + \mathcal{V}^*\mathcal{V}.$$

On input degree at most q, the first term has norm at most qt. Each middle term has norm at most $\sqrt{(a_0 + (q+1)t)(b_0 + (q+1)t)}$, and the last has norm at most $b_0 + (q+1)t$. These bounds also hold when $q = 0$, by the vanishing of the lowering maps on constants. Thus T raises degree by at most two and its restricted norm is at most

$$b_0 + (2q+1)t + 2\sqrt{(a_0 + (q+1)t)(b_0 + (q+1)t)}.$$

In forming $T^p(e_i \otimes 1)$, the input degrees are $0, 2, \dots, 2p-2$; the last display is bounded by $\mathcal{M}_p$ at each of these steps. Therefore

$$\mathbb{E}\,\mathrm{Tr}(T^{2p}) = \sum_{i=1}^{k} \|T^p(e_i \otimes 1)\|_{L^2}^2 \le k\mathcal{M}_p^{2p}.$$

Pointwise symmetry gives $\|T\|^{2p} \le \mathrm{Tr}(T^{2p})$. Finally, $\mathrm{off}(W^T W) = T - \mathrm{diag}(T)$ and $\|\,\mathrm{diag}(T)\| \le \|T\|$. This proves (1). □

3 Tensor blocks and the full matrix

We now apply the moment bound in a way that keeps its constants independent of d. We use the following form of [1, Theorem 1.4]. If Y is a tensor of order $q \ge 1$ with independent, centered,

unit-variance, K-subgaussian coordinates in its factors, and B is a deterministic linear map from $\mathbb{R}^{n^q}$ to a Euclidean space, then

$$\mathbb{P}\{|\|BY\| - \|B\|_{\mathrm{HS}}| \geq z\} \leq 2\exp\left(-\frac{a_K z^2}{qn^{q-1}\|B\|^2}\right), \qquad 0 \leq z \leq 2\|B\|_{\mathrm{HS}}. \tag{6}$$

The constant $a_K > 0$ depends only on K. Zero maps can be treated deterministically. We also use the upper tail in [2, Theorem 5.1.1]: for independent positive semidefinite $r_0 \times r_0$ matrices $Y_i \leq L_0 I$, with $\mu = \|\sum_i \mathbb{E}Y_i\| > 0$,

$$\mathbb{P}\left\{\left\|\sum_i Y_i\right\| \geq (1+\theta)\mu\right\} \leq r_0\left[\frac{e^\theta}{(1+\theta)^{1+\theta}}\right]^{\mu/L_0}, \qquad \theta \geq 0. \tag{7}$$

3.1 Projected blocks

Proposition 5. *Under the assumptions of Theorem 1, there is $\delta_K \in (0, 1/4)$, depending only on K, as follows. Put $h = N - m$ and $k = \lfloor \delta_K h \rfloor$. For each fixed k-subset J of columns, let P_J project orthogonally off the span of the columns outside J. For sufficiently large n, $1 \leq k \leq m$, and*

$$\mathbb{P}\{s_{\min}(P_J A_J) < \sqrt{\varepsilon N/8}\} \leq q_n, \qquad q_n \longrightarrow 0.$$

The bound q_n can be chosen independently of J and the coordinate laws.

Proof. For now let $\delta \in (0, 1/4)$ be a constant to be chosen using only K, and set $k = \lfloor \delta h \rfloor$. For fixed ε and sufficiently large n,

$$k \geq \delta h/2, \qquad 1 \leq k \leq m.$$

Fix J and condition on all columns outside J. Write $P = P_J$, $r = \operatorname{rank} P$, and

$$\beta = r/N, \qquad \gamma = k/N.$$

There are $m - k$ outside columns, even if some are dependent; hence

$$r \geq N - (m-k) = h + k, \qquad \beta \geq \varepsilon, \qquad \gamma \leq \delta\beta. \tag{8}$$

The projection is measurable without any rank assumption on those columns. If their matrix is C, it is the pointwise limit

$$P = \lim_{\lambda \downarrow 0}\left[I - C(C^T C + \lambda I)^{-1} C^T\right].$$

The singular-value decomposition verifies the limit. In particular, P is independent of all coordinates of columns in J, so conditioning means integration over the outside product law.

Let $R_0 = n^{d-1}$ and $b_i = x_{i1} \otimes \cdots \otimes x_{i,d-1} \in \mathbb{R}^{R_0}$. For $d = 1$, set $R_0 = 1$ and $b_i = 1$. After an orthogonal reordering of tensor coordinates, there are isometric embeddings $E_a : \mathbb{R}^{R_0} \to \mathbb{R}^N$ into n mutually orthogonal blocks such that

$$X_i = \sum_{a=1}^{n} \xi_{ia} E_a b_i,$$

where the ξ_{ia} are the coordinates of the last factor. Condition next on all b_i, $i \in J$, and define

$$U_i = \frac{[PE_1 b_i, \ldots, PE_n b_i]}{\sqrt{N}}, \qquad W = \frac{PA_J}{\sqrt{N}}. \tag{9}$$

We first establish, with high probability over the b_i,

$$\max_{i\in J}\|b_i\|^2 \le 2R_0, \qquad \left\|\sum_{i\in J} b_i b_i^T\right\| \le 2k, \qquad \max_{i\in J}\|U_i\|_{\mathrm{HS}}^2 \le 9\beta/4. \tag{10}$$

For $d \ge 2$, apply (6) to the identity on order $d-1$ tensors, with $z = (\sqrt{2}-1)\sqrt{R_0}$, for the first bound. For the second, truncate separately in each column:

$$Y_i = b_i b_i^T \mathbf{1}_{\{\|b_i\|^2 \le 2R_0\}}.$$

These matrices are independent, $0 \le Y_i \le 2R_0 I$, and $\mathbb{E}Y_i \le I$. To see the last assertion, note that $\mathbb{E}b_i b_i^T = I$: for tensor multi-indices, the covariance factors into scalar covariances and equals one on the diagonal and zero otherwise. This uses independence and unit variance, with no equality of laws. Let $\mu = \|\sum_i \mathbb{E}Y_i\| \le k$. If $\mu = 0$, positivity and a zero expected trace give $\sum_i Y_i = 0$ almost surely. Otherwise take $\theta = 2k/\mu - 1 \ge 1$ in (7). The logarithm of the bracketed probability factor is

$$\frac{2k - \mu - 2k\log(2k/\mu)}{2R_0}.$$

This is increasing in μ for $0 < \mu \le k$, and its value at k is $-(2\log 2 - 1)k/(2R_0)$. Therefore

$$\mathbb{P}\left\{\left\|\sum_i Y_i\right\| > 2k\right\} \le R_0 \exp\left(-\frac{(2\log 2 - 1)k}{2R_0}\right). \tag{11}$$

On the first event in (10), $\sum_i Y_i = \sum_i b_i b_i^T$.

For the third bound in (10), put $T_P = \sum_a E_a^T P E_a$. Then $0 \le T_P \le nI$ and $\operatorname{Tr} T_P = r$. Apply (6) to $T_P^{1/2}$ and b_i, with $z = \sqrt{r}/2$. Since $\|U_i\|_{\mathrm{HS}}^2 = b_i^T T_P b_i/N$, this gives the desired upper bound. All these applications are uniform in the fixed projection P. For $d = 1$, the three bounds hold deterministically: $\|b_i\|^2 = 1$, $\sum_i b_i b_i^T = k$, and $\|U_i\|_{\mathrm{HS}}^2 = \beta$.

On (10), the parameters of Lemma 4 for these U_i satisfy

$$a_0 \le 9\beta/4, \qquad b_0 \le 2\gamma, \qquad t \le 2/n. \tag{12}$$

For the middle inequality,

$$\sum_i U_i U_i^T = \frac{P(I_n \otimes \sum_i b_i b_i^T)P}{N}.$$

For the last, the unprojected columns $E_a b_i$ are orthogonal, so $\|U_i\| \le \|b_i\|/\sqrt{N}$.

The off-diagonal Gram matrix in (9) is a quadratic polynomial to which Lemma 3 applies. Explicitly, set $C_{(i,a)} = PE_a b_i e_i^T/\sqrt{N}$. For $\alpha = (i,a)$, $\lambda = (j,b)$, take

$$H_{\alpha\lambda} = \begin{cases} (C_\alpha^T C_\lambda + C_\lambda^T C_\alpha)/2, & i \ne j, \\ 0, & i = j. \end{cases}$$

These coefficients are symmetric as matrices and in their two indices, and $\operatorname{off}(W^T W) = \sum_{\alpha,\lambda} H_{\alpha\lambda}\xi_\alpha\xi_\lambda$. The last-factor coordinates retain their independent centered laws after the conditioning. Lemmas 3 and 4 therefore give

$$\|\operatorname{off}(W^T W)\|_{L^{2p}} \le 4(16K)^2 k^{1/(2p)} \mathcal{M}_p,$$

where this moment is conditional on $P, (b_i)$. Choose $p = \lceil \log N \rceil$, and put $\rho = 2(2p+1)/n$ and $B_K = 4(16K)^2\sqrt{e}$. Since $k \le N$, (12) bounds the last display by

$$B_K \left[2\gamma + 2\rho + 2\sqrt{(9\beta/4 + \rho)(2\gamma + \rho)} \right]. \tag{13}$$

Choose $\delta = \delta_K \in (0, 1/4)$ so that

$$B_K[2\delta_K + 3\sqrt{2\delta_K}] < 1/(16e). \tag{14}$$

This choice uses only K. Dividing (13) by β, using (8), and setting $\eta = \rho/\beta$ gives the upper bound

$$B_K \left[2\delta_K + 2\eta + 2\sqrt{(9/4 + \eta)(2\delta_K + \eta)} \right].$$

At $\eta = 0$, this is less than $1/(16e)$. By continuity it is at most $1/(8e)$ for $0 \le \eta \le \eta_K$, for some $\eta_K > 0$ depending only on K. Since $\rho/\varepsilon \to 0$ and $\beta \ge \varepsilon$, this holds for all sufficiently large n. Thus, on (10), Markov's inequality gives

$$\mathbb{P}\{\| \operatorname{off}(W^T W)\| > \beta/8 \mid P, (b_i)\} \le e^{-2p} \le N^{-2}. \tag{15}$$

For the actual diagonal entries, condition only on P. Apply (6) at order d, with $B = P$, $\|P\|_{\mathrm{HS}} = \sqrt{r}$, and $z = \sqrt{r}/2$. A union bound gives

$$\min_{i \in J} \frac{\|PX_i\|^2}{N} \ge \beta/4 \tag{16}$$

except with probability at most $2k \exp(-a_K \beta n/d)$, after decreasing a_K by an absolute factor. There is no need for independence between (16) and the event in (15). Their intersection implies $\lambda_{\min}(W^T W) \ge \beta/8$, and hence

$$s_{\min}(PA_J) \ge \sqrt{\beta N/8} \ge \sqrt{\varepsilon N/8}.$$

Combining the three tensor concentration bounds, the covariance tail and (15), the failure probability is at most

$$q_n = 6N \exp(-a_K \varepsilon n/d) + R_0 \exp(-c_0 \delta_K \varepsilon n) + N^{-2}, \qquad c_0 = (2\log 2 - 1)/4, \tag{17}$$

for sufficiently large n, with a further decrease of a_K depending only on K. Here $k/R_0 \ge \delta_K \varepsilon n/2$. The first term tends to zero because

$$\frac{d \log n}{\varepsilon n/d} = \frac{d^2 \log n}{\varepsilon n} \longrightarrow 0,$$

the second because $\log R_0 = (d-1)\log n = o(n)$, and the third because $N \to \infty$. Also $\rho/\varepsilon = O(d \log n/(\varepsilon n)) \to 0$, as used above. For $d = 1$, the auxiliary events in (10) are deterministic, so the same upper bound remains valid. Every bound is uniform in J, the conditioned projection, and the permitted laws. □

3.2 Subset averaging

We conclude with a deterministic implication for any matrix. For a fixed unit vector $v \in \mathbb{R}^m$, let J be uniform among the k-subsets of $\{1, \dots, m\}$, and write

$$Z = \|v_J\|^2, \qquad t_0 = k/m.$$

Then $0 \le Z \le 1$ and $\mathbb{E}_J Z = t_0$. It follows that

$$\mathbb{P}_J\{Z \ge t_0/2\} \ge \frac{t_0}{2 - t_0} \ge t_0/2, \tag{18}$$

by bounding $\mathbb{E}_J Z$ above by $t_0/2 + (1 - t_0/2)\mathbb{P}_J\{Z \ge t_0/2\}$.

Proof of Theorem 1. Take δ_K from Proposition 5, and set

$$c_K = \sqrt{\delta_K}/8, \qquad k = \lfloor \delta_K (N - m) \rfloor.$$

For sufficiently large n,

$$t_0 = k/m \ge \delta_K \varepsilon/2.$$

For any realization of A with $s_{\min}(A) < c_K \varepsilon \sqrt{N}$, choose a unit vector v with $\|Av\| < c_K \varepsilon \sqrt{N}$. If $Z \ge t_0/2$ and the projected block for J obeyed $s_{\min}(P_J A_J) \ge \sqrt{\varepsilon N/8}$, then

$$\|Av\| \ge \|P_J A v\| = \|P_J A_J v_J\| \ge \sqrt{\varepsilon N/8}\sqrt{t_0/2} \ge \sqrt{\delta_K/32}\,\varepsilon\sqrt{N},$$

contradicting the choice of c_K. By (18), a fraction at least $t_0/2$ of all subsets must therefore have a bad projected block. This is a pointwise statement for each matrix and needs no measurable selection of v. Integrating this finite fraction over the law of A, and using Proposition 5, gives

$$\mathbb{P}\{s_{\min}(A) < c_K \varepsilon \sqrt{N}\} \le \frac{2m}{k} \mathbb{P}_{A,J}\{s_{\min}(P_J A_J) < \sqrt{\varepsilon N/8}\} \le \frac{4}{\delta_K \varepsilon}\, q_n \longrightarrow 0.$$

The choice of c_K uses only K, and (17) applies to every admissible order and coordinate-law sequence. This proves the stated fixed-ε convergence. □

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